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FADIL NAUFAL WAHAS

EDUCATION

Universitas Indonesia, Depok – Master of Engineering (M.T.) in Mechanical Engineering
Aug 2023 - Feb 2026

Specialization of Conversion and Conservation of Energy
Master's Thesis Topic: Two-phase Turbulent CFD Simulation of a Nuclear Reactor Cooling System

Universitas Gadjah Mada, Sleman – Bachelor of Engineering (S.T.) in Nuclear Engineering
Aug 2017 - Aug 2022

Specialization of Nuclear Energy Technology
Undergraduate Thesis: "Analysis of the Temperature Reactivity Coefficient and the Power Peaking Factor of Micro Reactor Heat Pipe (MRHP) with Uranium Carbide Using OpenMC"

WORK EXPERIENCES

P2M Universitas Indonesia, Depok – Associate Consultant
May 2026 - Present

- Assessed, by field survey, the mechanical, electrical, and plumbing (MEP) system of Indonesian bank buildings, including offices, data centers, and vault facilities.
- Prepared and presented a comprehensive MEP system survey report to support the development of the maintenance guidebooks.
- Composed the preventive and corrective maintenance guidebook of the plumbing system.

Universitas Indonesia, Depok – Trainer of Python Programming
Feb 2024 - Feb 2025

- Led 20 participants on basic training of Python programming.
- Reviewed and debugged participant's code on 5 modules.
- Contributed to the new syllabus materials including basic Artificial Neural Network and Machine Learning.
- Created a new plagiarism detection protocol for assignments using Python script.
- Examined and graded the participant's work and report on each module of the training.

National Nuclear Energy Agency of Indonesia (BATAN), South Tangerang –
Intern of Reactor Operation Planning Division at Center of Multipurpose Reactor (PRSG)
Jan 2020 - Feb 2020

- Participated in pre and post-processing of reactor operations and Topaz irradiation process with 20 work hours inside the reactor building per week.
- Analyzed the reactivity changes in Topaz irradiation loading/unloading as a scientific report.
- Improved the reactor Safety Analysis Report data with 2 findings.
- Presented the results of analysis as an internship completion requirement.

Universitas Gadjah Mada, Sleman – Laboratory Assistant of Computer Programming
Jan 2019 - Jul 2019

- Led 15 students on basic C++ programming language on Computer Programming practicum.
- Reviewed and debugged student code on 8 modules.
- Managed student data and communication flow among 3 classrooms.
- Examined and graded the student's work and report on each module of the practicum.

ORGANIZATIONAL AND VOLUNTEERING EXPERIENCES

KKN-PPM UGM, Yogyakarta – Volunteer
May 2020 - Aug 2020

- Initiated a seminar addressing the technology literacy issues for 25 people in one district of the village.
- Engineered a concept of sanitation station during Covid-19 pandemic for 50% less cost than a commercial one.
- Collaborated as a team composed of 20 students from different areas of expertise.
- Created technical report and documentation on the execution of volunteering activities.

Purna Paskibraka Indonesia, Jakarta – Member
Aug 2015 - Present

- Organized and executed a leadership training camp with 150 attendees in the events division.
- Functioned in youth activities, 2 social events, 2 volunteering, and 1 national and 1 international sporting events.

TRAININGS AND CERTIFICATIONS

C++ Programming – Saylor Academy
Jun 2026 | Certificate

- Fundamentals of C++ programming language and data structure
- Implemented object-oriented programming
- File I/O operations and directives

Risk Management In Technical Cooperation Projects – International Atomic Energy Agency (IAEA) Open-LMS
Sep 2023 | Certificate

- Risk management system in technical projects
- Risk assessment, identification, and response
- Risk monitoring and reporting

Scientific Computing with Python – freeCodeCamp.org
Jan 2023 | Certificate

- Fundamentals of Python, data structures, and object-oriented programming
- Scientific computation for engineering
- Data analysis, visualization, and database system

IAEA Safety Standards Overview – International Atomic Energy Agency (IAEA) Open-LMS
Nov 2022 | Certificate

- Understanding of IAEA's Safety Standards, principles, objectives, and its history
- Safety fundamentals, requirements, and guides (IAEA Safety Publication Series)
- IAEA Safety Standards development, establishment, evaluation, access, and implementation

SKILLS AND LANGUAGES

Technical Skills:

- Engineering in the fields of energy, mechanical, fluid dynamics, and nuclear
- Engineering system optimization
- Process engineering and scientific calculation using Python and MATLAB
- Design and safety of nuclear reactor using OpenMC, MCNP, and SCALE
- Fluid mechanics and modeling of Computational Fluid Dynamics (CFD) using OpenFOAM
- Technical design: Detailed Engineering Drawing (DED), Computer Aided Design (CAD), Piping and Instrumentation Diagram (P&ID), and As-Built Drawing (ABD) documentations
- Scientific and technical writing and documentation with office programs
- Python and C++ programming languages
- Data analysis, Artificial Neural Network (ANN) modeling, and Machine Learning
- Linux environment, GitHub repositories, and code distribution and development

Languages: Professional working proficiency in English; Fluent in Indonesian